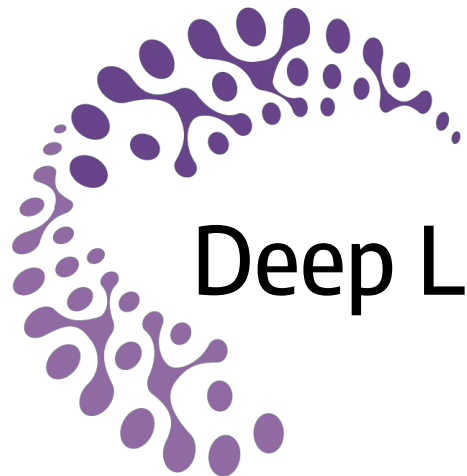




Deep Learning

CSE0613409

Machine Vision and Motion Analysis

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(MMF)

Department of Computer Science and Engineering

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- Deep Learning definition
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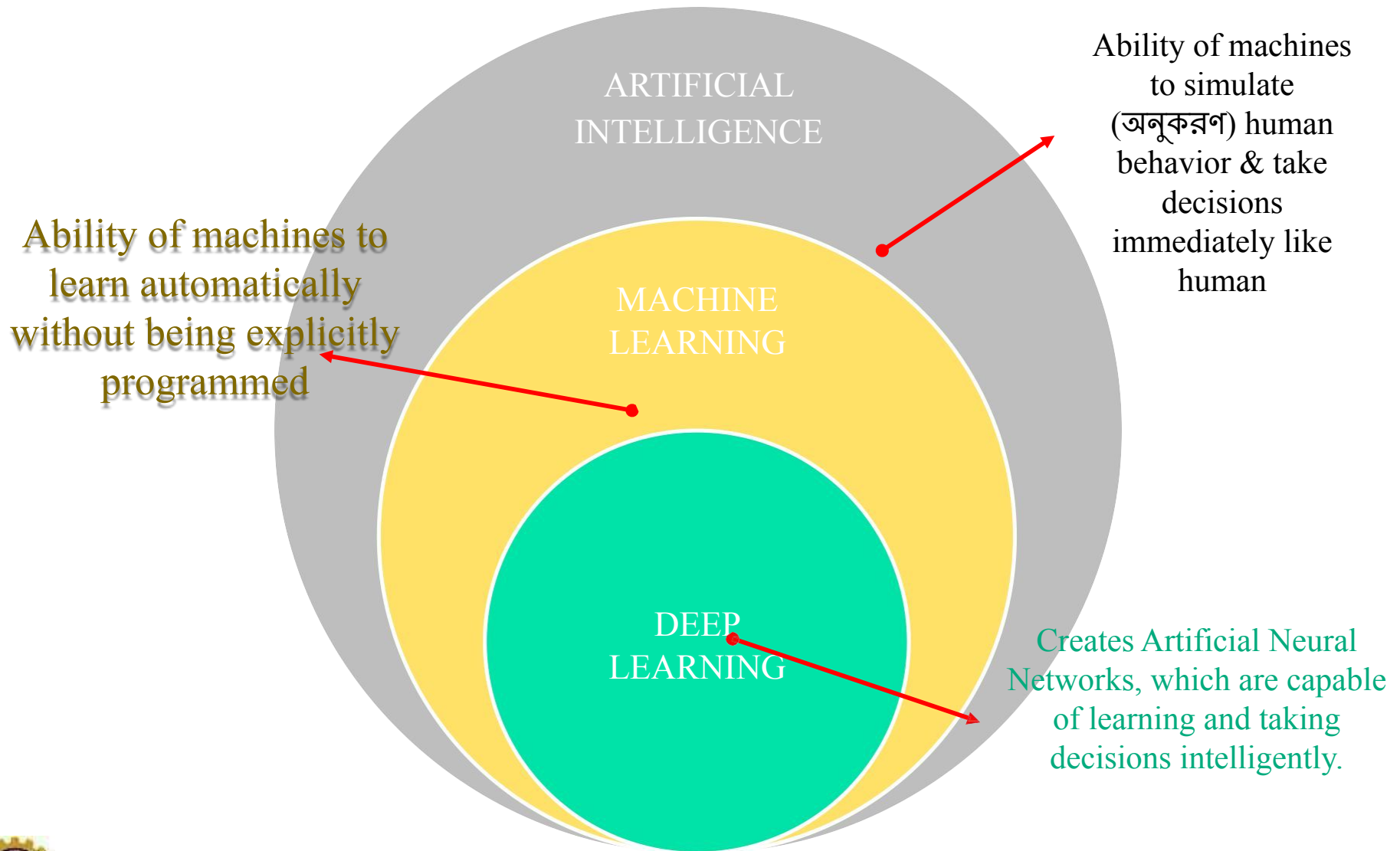


Objectives

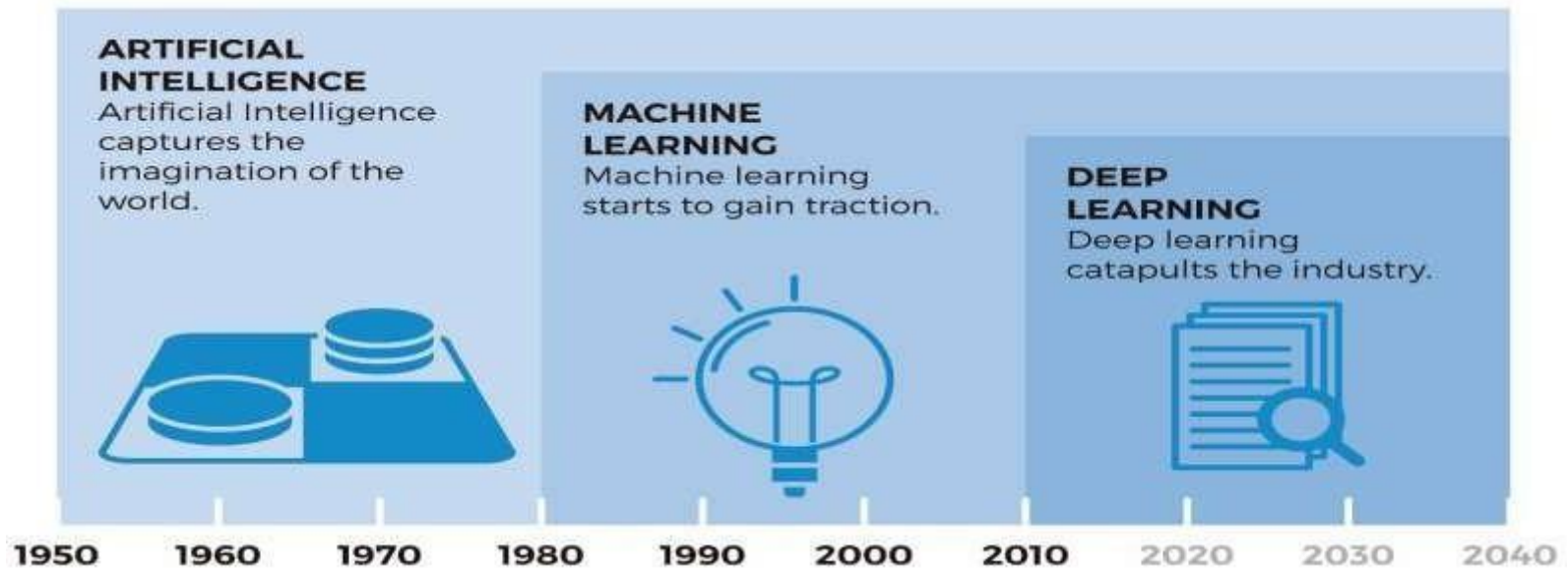
- ❑ To give ideas about artificial intelligence, machine learning and deep learning.
- ❑ To provide an description about deep learning and difference between deep learning and machine learning.
- ❑ To provide an description of activation functions.
- ❑ To provide an description of Neural Networks with use-case.
- ❑ To provide an description of ANN.
- ❑ To provide an description of CNN with use-case.
- ❑ To provide ideas about different layers in CNN.



Introduction



Evaluation of AI,ML,DL

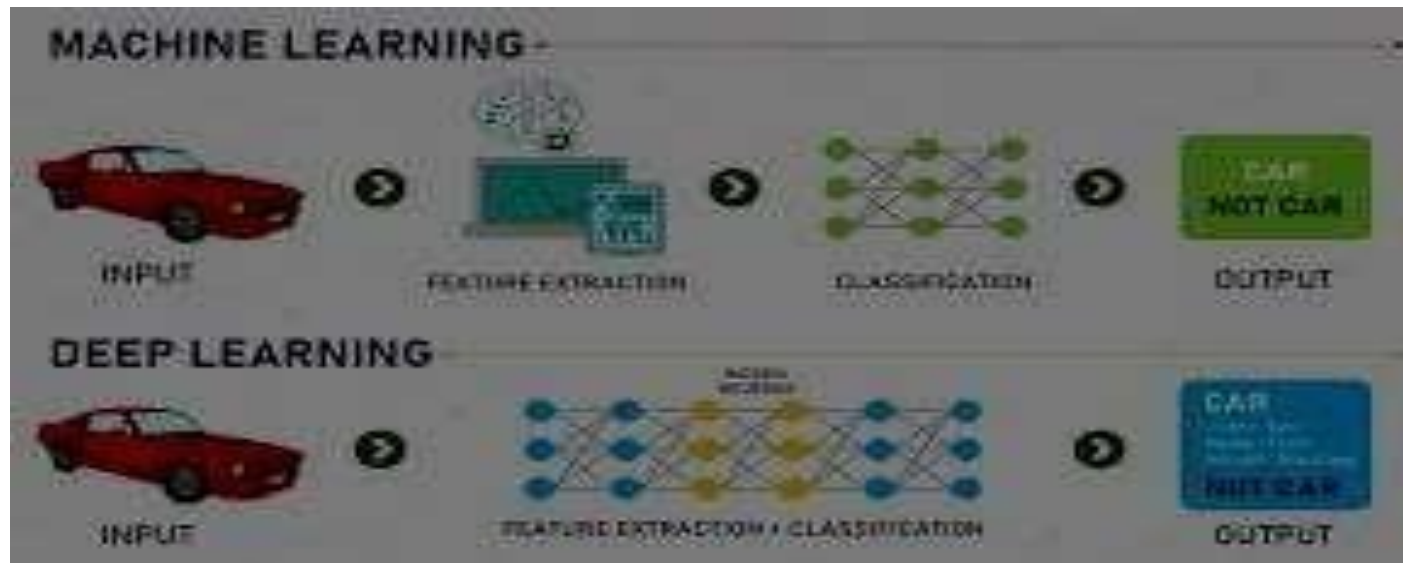


Deep Learning Definition

What is Deep Learning?

Deep Learning is a class of machine learning algorithms in artificial intelligence that uses multiple layers to progressively extract high level features from the raw data. It is entirely concerned with algorithms inspired by the structure and function of artificial neural networks(ANN) which are inspired by the human brain.

Machine Learning VS Deep Learning:



Deep Learning(Continued)

“Deep” in Deep Learning:

Deep in deep learning stands for **the idea of successive layer of representation**. Number of layers contribute in a model is called the depth of the model. **Machine learning model mainly focus learning only one or two layers of representation of data. But modern deep learning involves tens or even hundreds of successive layers of representations and they are all learned automatically from exposure to training data.** In deep learning, these layered representations are (almost always) learned via models called neural networks, structured in literal layers stacked on top of each other.

Why deep learning is so famous?

Deep learning is famous for many reasons. The reasons are given below:

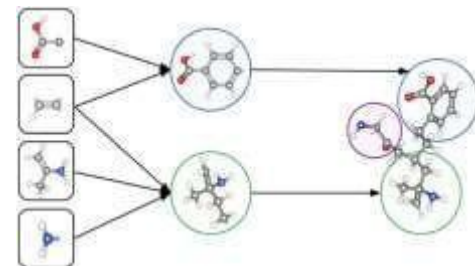
1. **Hardware and software advancement** – Deep learning is highly amenable to parallelization on GPUs and cloud facilities.
2. **Data growth** – Deep-learning models can be trained on additional data without restarting from scratch, making them viable for continuous online learning
3. **Algorithm** – Deep learning removes the need for feature engineering, replacing complex, brittle, engineering-heavy pipelines with simple, end-to-end trainable models



Applications of Deep Learning

Toxicity detection for different chemical structures

Deep learning method is very efficient, where experts used to take decades of time to determine the toxicity of a specific structure, but with deep learning model it is possible to determine toxicity in very less amount of time.



Automatic Machine Translation

With the help of deep learning, Google has totally reformed the approach to machine translation in its Google Translate.



Hallucination or Sequence generation

Creating new footage by observing different video games, learning how they work and replicate them using deep learning techniques like recurrent neural networks. Deep learning hallucinations can generate High-resolution images by using low-resolution images.



Applications of Deep Learning(Continued)

Speech Recognition

As a human recognize speech understands it and responds accordingly, the same way deep learning model is enhancing the capabilities of computers so that they can understand how humans do react to different speeches. .



Fraud Detection

With deep learning models, it is also possible to find out which product and which markets are most susceptible to fraud and provide or extra care in such cases.

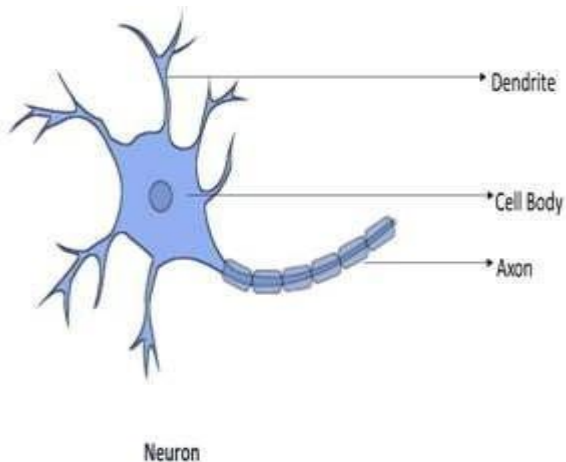


Facial Recognition and Automatic Tagging

In Facebook people in a certain images are recognized and can be automatically tagged. This can be possible only for the concept of deep learning.



How Deep Learning Works



Biological	Artificial
Dendrites	Inputs
Cell Nucleus	Nodes
Synapse	Weights
Axon	Outputs

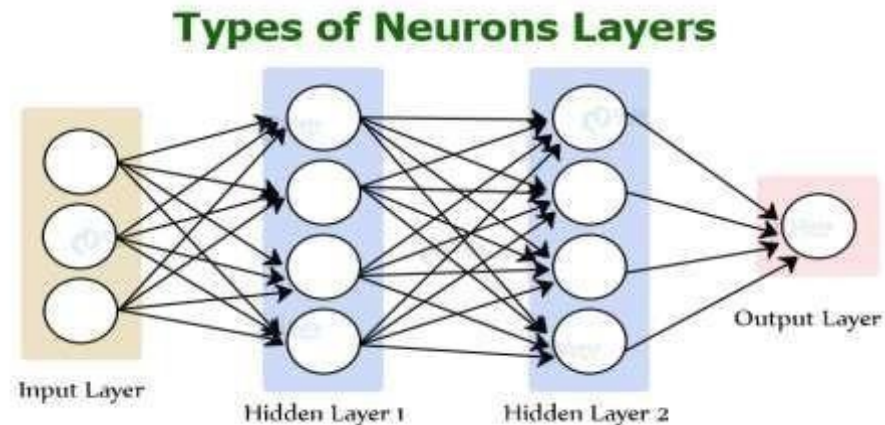
- ❖ Deep Learning studies **the basic unit of a brain called a brain cell or a neuron.** Parts of neuron is given below:
Dendrite: Receives signals from other neurons
Cell Body: Sums all the inputs
Axon: It is used to transmit signals to the other cells
- ❖ As we know that our brain consists of multiple connected neurons called neural network, we can also have **a network of artificial neurons called perceptrons to form a Deep neural network.**



How Deep Learning Works(Continued)

The neural network sorts the unlabeled data according to the similarities of the data. That's the idea behind a deep learning algorithm. Any Deep neural network will consist of three types of layers:

1. The Input Layer
2. The Hidden Layer
3. The Output Layer



Input Layer: It receives the input data from the observation. This information breaks into numbers and the bits of binary data that a computer can understand. Variables need to be either standardized or normalized to be within the same range.

Hidden Layer: It performs mathematical computations on input data. It does the non-linear processing units for feature extraction and transformation. Each following layer utilizes the output of the preceding layer as input.

Output Layer: It is responsible for computations and transferring information from the network to the outside world.

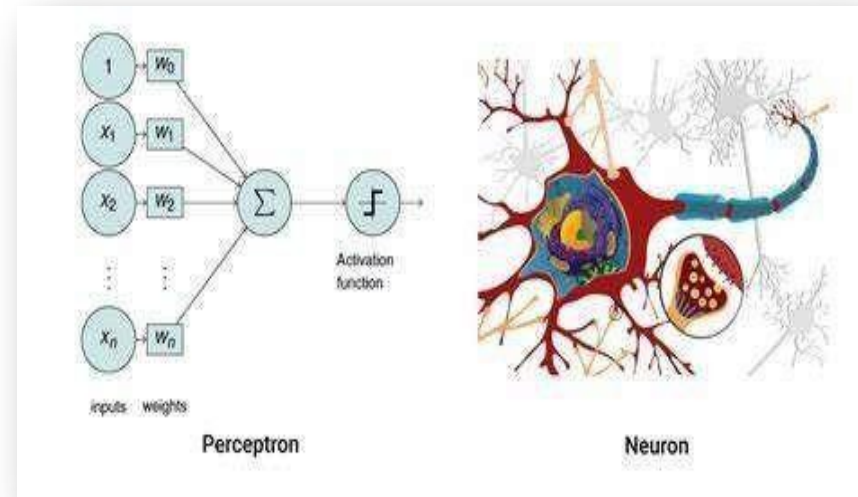


How Deep Learning Works(Continued)

An artificial neuron (perceptron) :

An Artificial Neuron or a Perceptron is a simple model of a biological neuron in an artificial neural network. It models a neuron which has a set of inputs, each of which is given a specific weight. It receives n inputs and then sums those inputs, applies a transformation and produces an output. It has two functions:

1. Summation
2. Transformation(Activation)



Weight: The weight shows the effectiveness of a particular input. More the weight of input, more it will have an impact on the neural network.

Bias: Bias is an additional parameter in the Perceptron which is used to adjust the output along with the weighted sum of the inputs to the neuron which helps the model in a way that it can fit best for the given data.

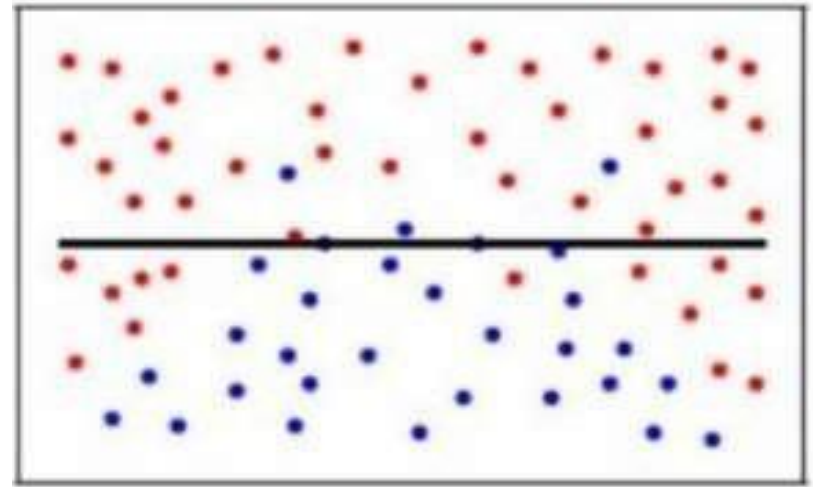


Bias Definition

Bias: Bias is the error that is introduced by the model's prediction and the actual data.

Bias = Predicted – Actual

- High Bias means the model has created a function that fails to understand the relationship between input and output data.
- Low Bias means the model has created a function that has understood the relationship between input and output data.



The input variables be X and a target variable Y . We map the relationship between the two using a function f .

$$Y = f(X) + e$$

‘ e ’ is the error that is normally distributed. The aim of our model $f(x)$ is to predict values as close to $f(x)$ as possible. Here, the Bias of the model is:

$$\text{Bias}[f'(X)] = E[f'(X) - f(X)]$$

Since, it does not learn the training data very well, it is called **Underfitting**.

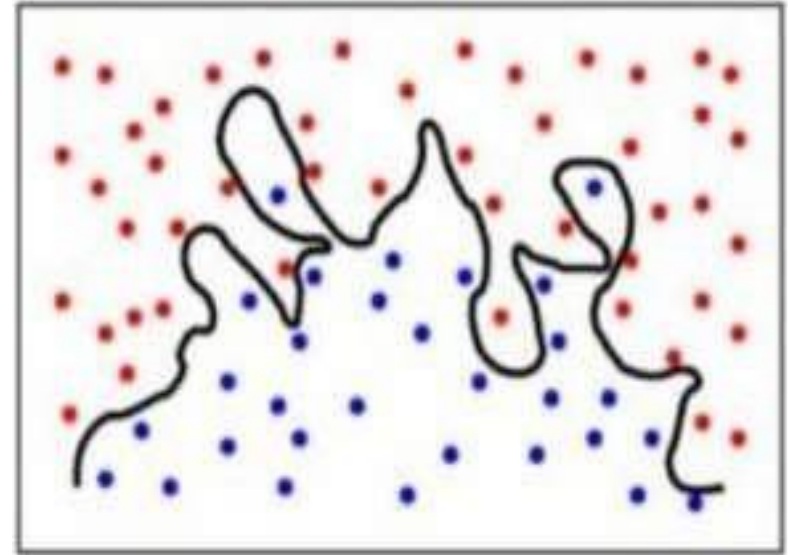
Underfitting : If a model fails to show satisfactory accuracy during the training phase itself it means the model is underfitting.



Variance Definition

Variance : The variance of the machine learning model is the amount by which its performance varies with different data set..

- Low variance means, the performance of the machine learning model does not vary much with the different data set.
- High variance means, the performance of the machine learning model varies considerably with different data set..



Mathematically, the variance error in the model is:

$$\text{Variance}[f(x)] = E[X^2] - E[X]^2$$

Since in the case of high variance, the model learns too much from the training data, it is called **overfitting**.

Overfitting : If a model is showing high accuracy during the training phase but fails to show similar accuracy during the testing phase it indicates overfitting.



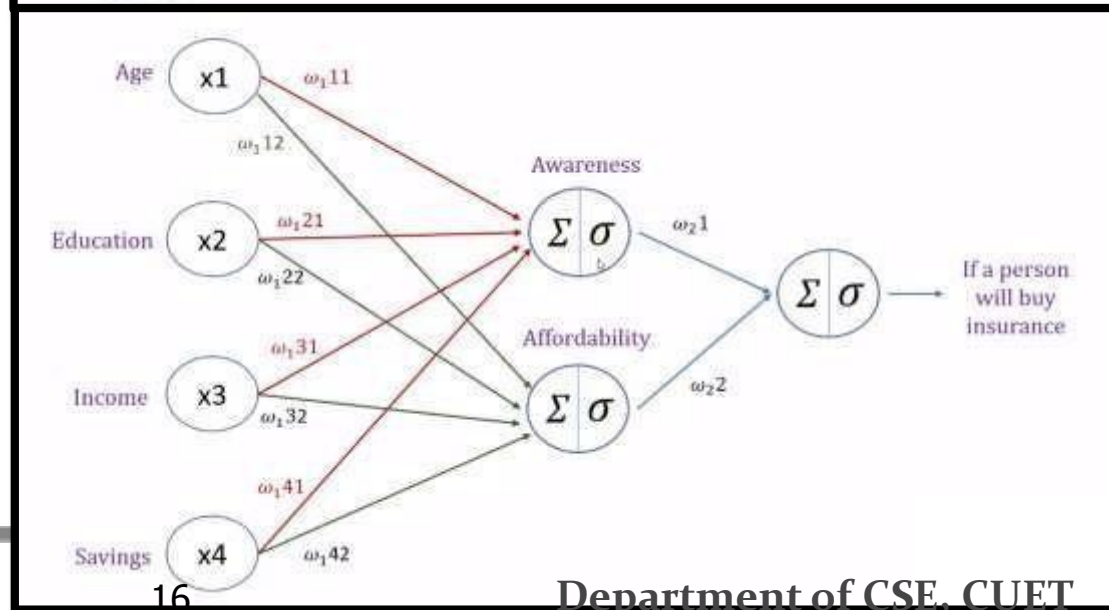
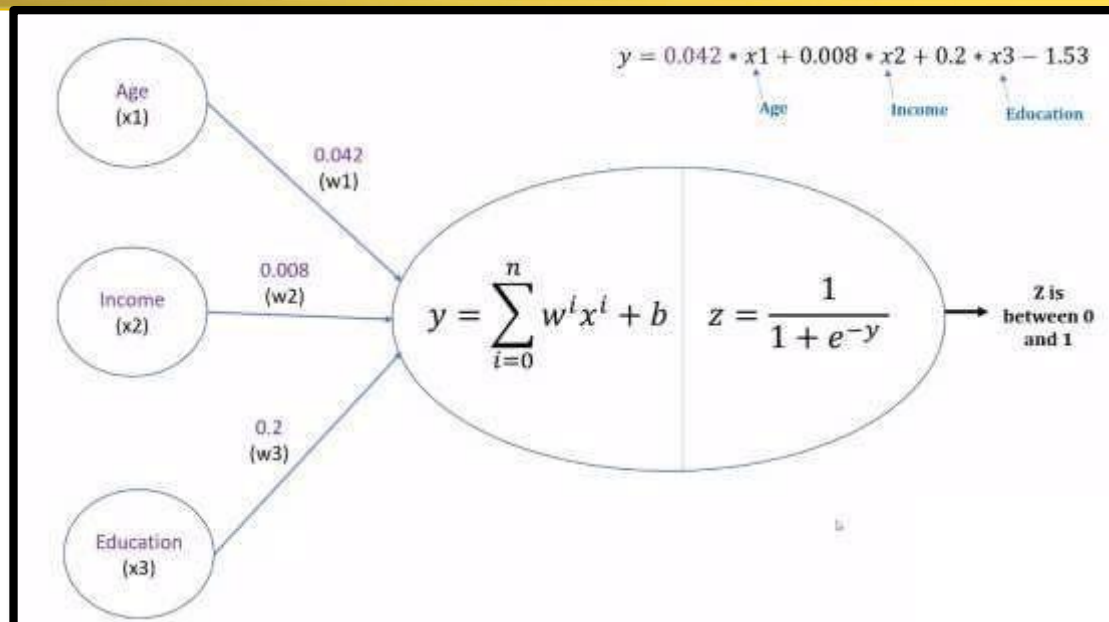
Activation Function

Activation Function:

An activation function is a function that is added into an artificial neural network in order to help the network learn complex patterns in the data [i.e. firing(1) or not firing (0)]. Activation functions add non-linearity to the model.

There are many functions that are used as Activation Functions, like:

- Step Function
- Linear Function
- Sigmoid Function
- Tanh Function
- ReLU Function (Rectifier Linear Unit)
- Leaky ReLU Function

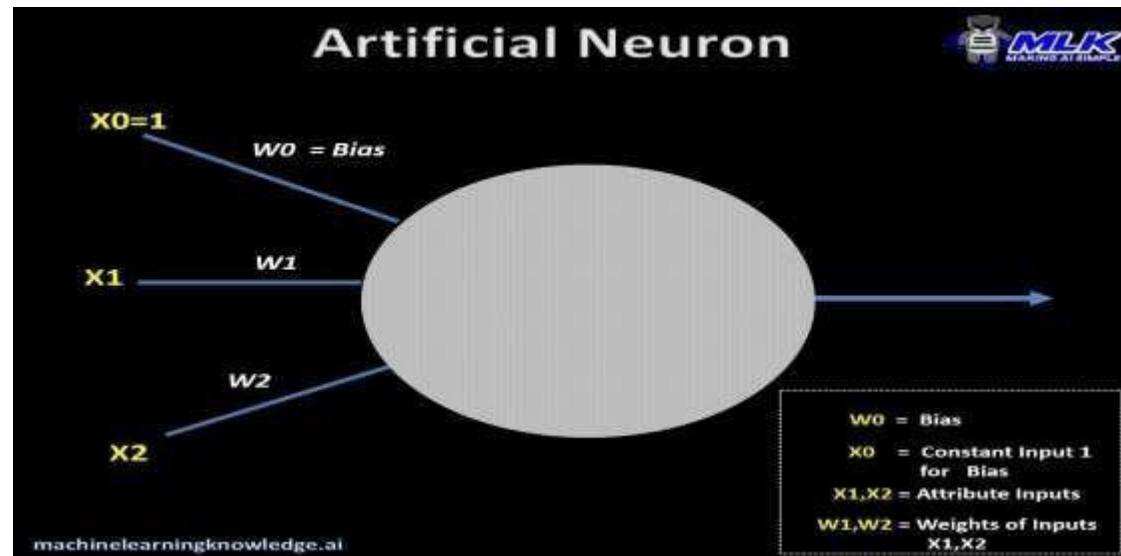


Necessary of Activation Function

Necessary of Activation Function:

An artificial neuron without an activation function will just produce the sum of dot products between all inputs and their weights.

Input : X_0, X_1, X_2 then Output = $W_0X_0 + W_1X_1 + W_2X_2$ [without activation function].



So, without activation function the network will produce a linear output and there is not any necessary of hidden layer. But, in the real world hardly any data is so simple that it can be represented with just a linear mapping function. The activation function added this non-linearity property especially in the neurons of hidden layers to the network. Thus we can say that Neural networks are used to implement complex functions, and non-linear activation functions enable them to approximate arbitrarily complex functions. Without the non-linearity introduced by the activation function, multiple layers of a neural network are equivalent to a single layer neural network.

Step Function

Step Function:

This is a very intuitive activation function which only enables or disables artificial neuron. It can only produce two values 0 or 1. If it is 0 it means neuron has not fired, if it is 1 it means neuron has fired. It produces **output 1 only when the sum of dot products of inputs & weights is more than a threshold value, else it will be 0.**

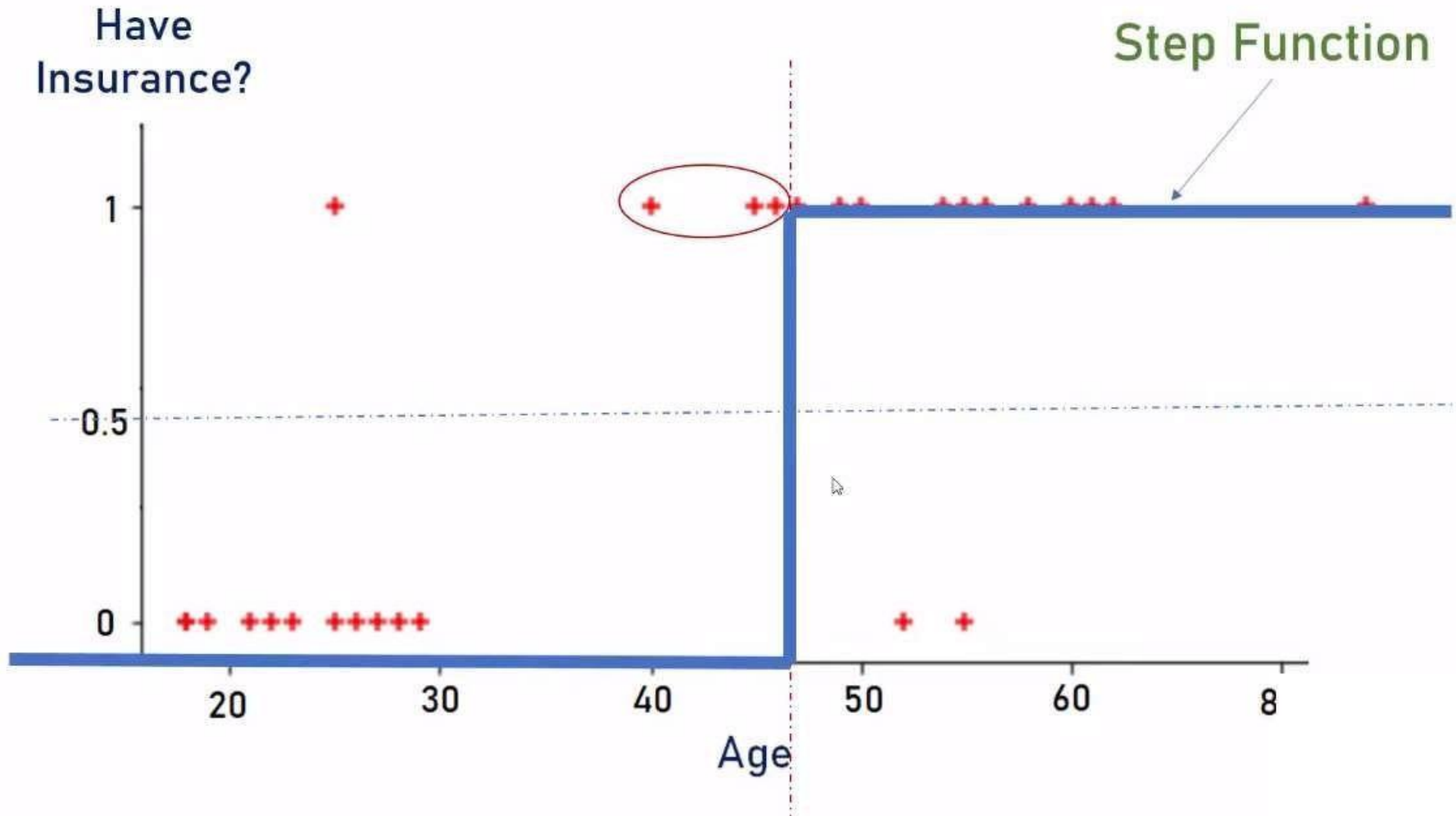
Equation:

$$u(t) = \begin{cases} 0 & \text{for } t < 0 \\ 1 & \text{for } t \geq 0 \end{cases}$$

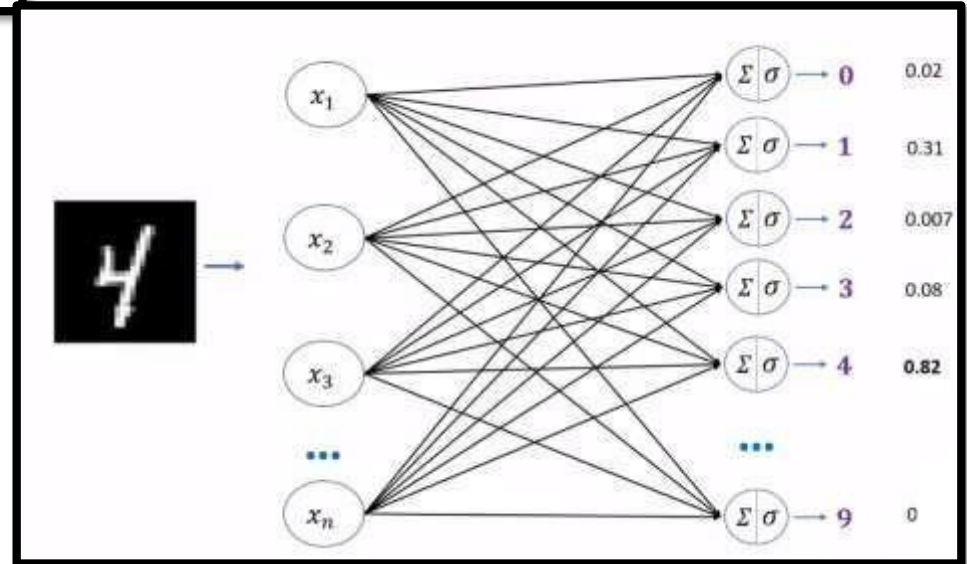
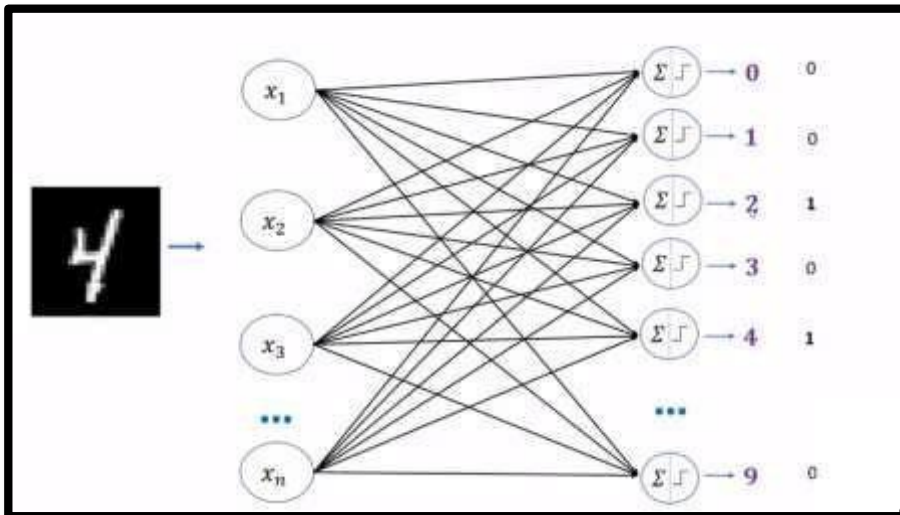
Drawbacks :

1. The main drawback of step function is as in below image where you want to classify multiple such neurons into Classes one , Classes two & Classes three etc. **so what will happen If more than one neuron is activated, all these neurons will output of 1.**
2. When **outlier exists then step function might misclassify the problem (i.e. hard to make decisions).**

Step Function



Step Function VS. Sigmoid Function

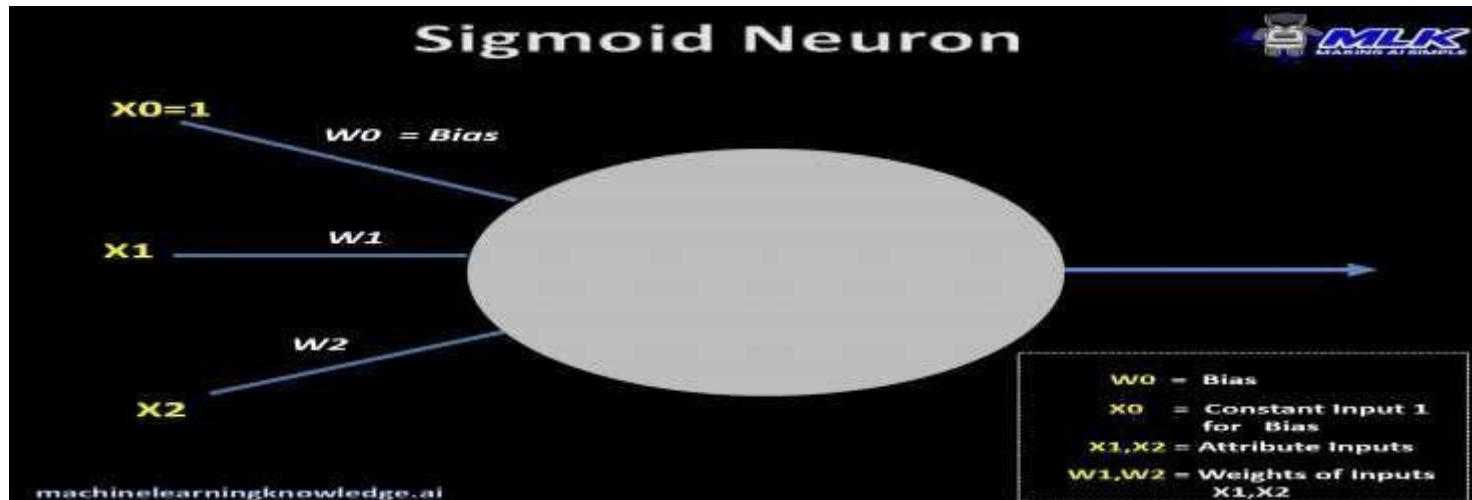


Sigmoid Function

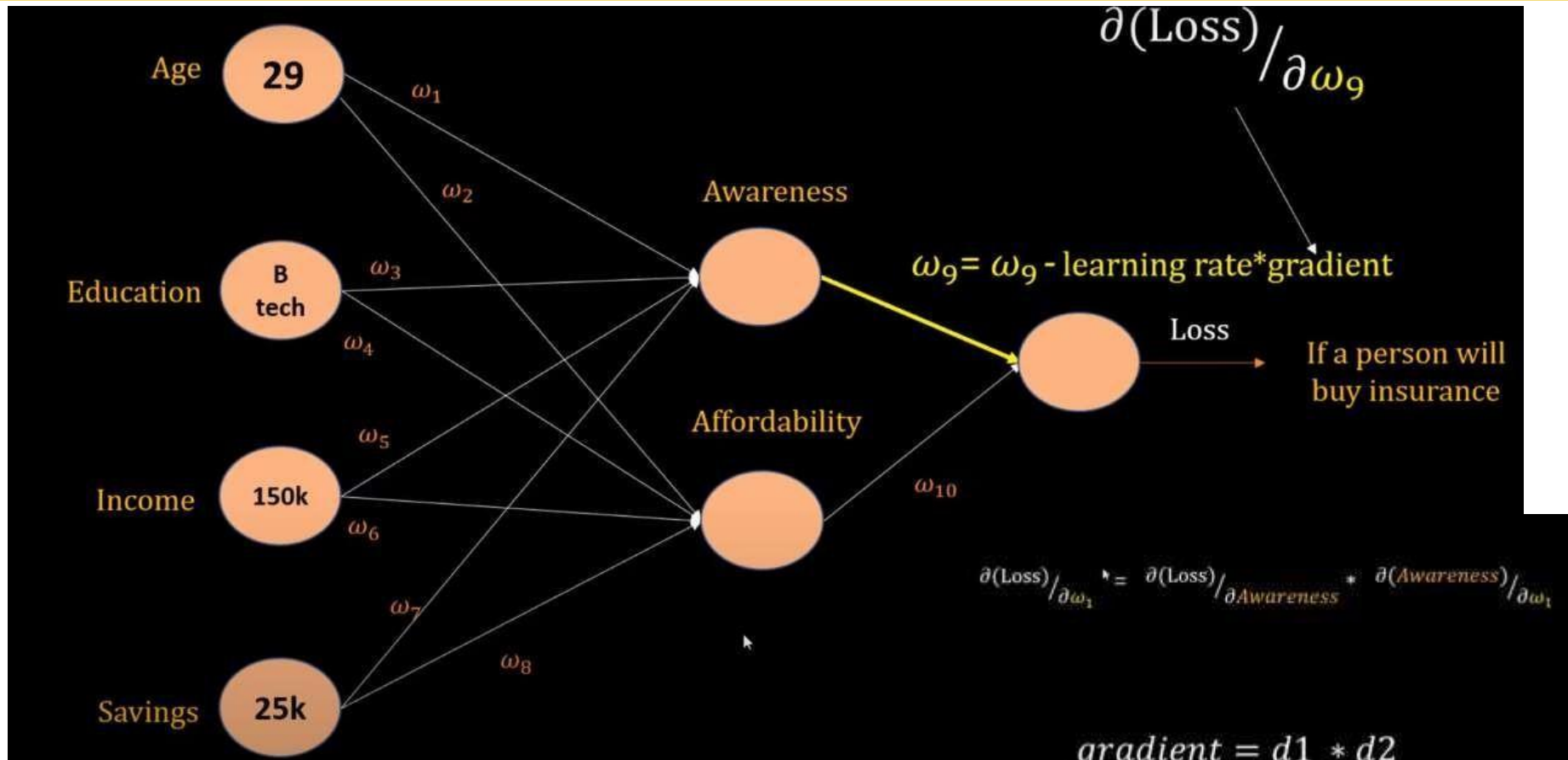
Sigmoid Function : The sigmoid function is S Shaped and is smooth. The range of output is between 0 and 1 which is usually interpreted as probability if used in the output layer. This function is both non-linear and is differentiable.

Equation:

$$f(x) = \frac{1}{1 + e^{-(x)}}$$



Sigmoid Function(Drawbacks) : Vanishing Gradient



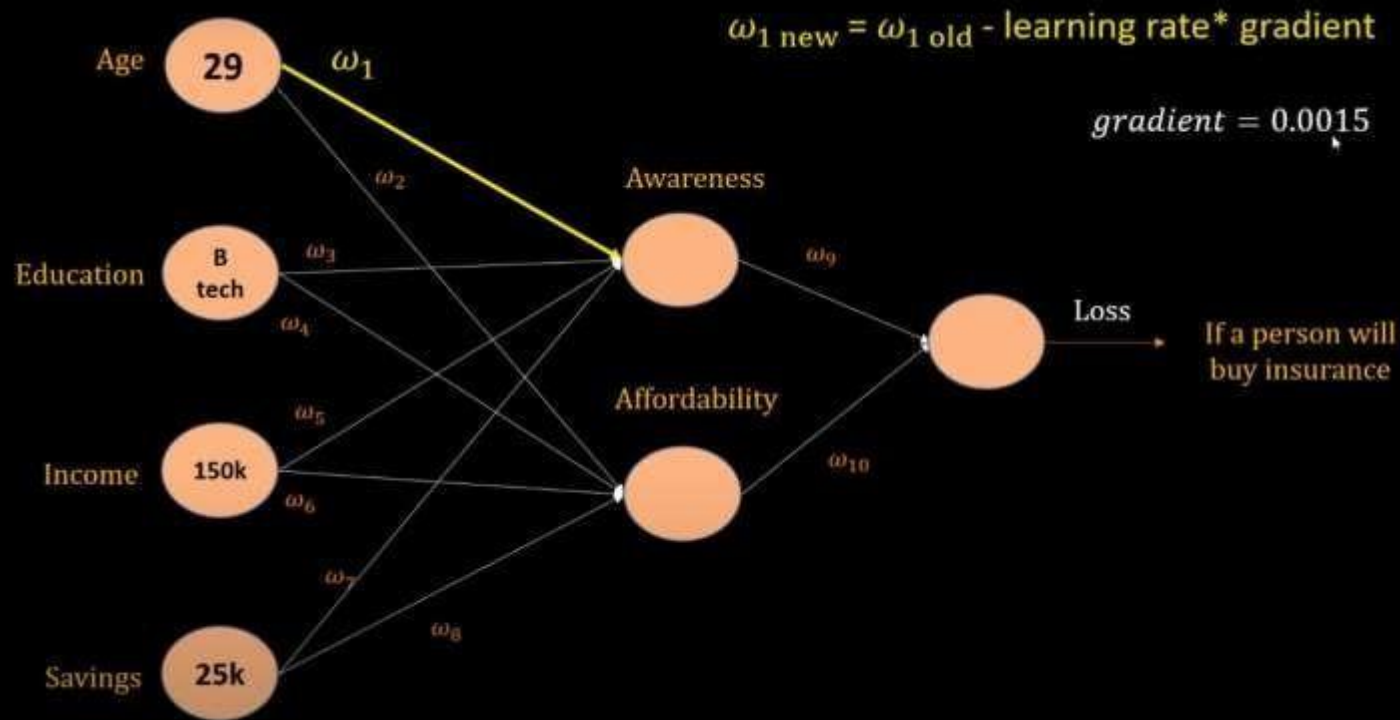
$$\text{gradient} = d1 * d2$$

$$\text{gradient} = 0.03 * 0.05$$

$$\text{gradient} = 0.0015$$

Department of CSE, CUET





As number of hidden layers grow, gradient becomes very small and weights will hardly change . This will hamper the learning process.

Vanishing Gradients

$$\frac{\partial(\text{Loss})}{\partial \omega_1} = \frac{\partial(\text{Loss})}{\partial \text{Awareness}} * \frac{\partial(\text{Awareness})}{\partial \omega_1}$$

$$\text{gradient} = d1 * d2$$

$$\text{gradient} = 100 * 500$$

$$\text{gradient} = 50000$$

When individual derivatives are large, the final derivative will also become huge and weights would change drastically.

Exploding Gradients



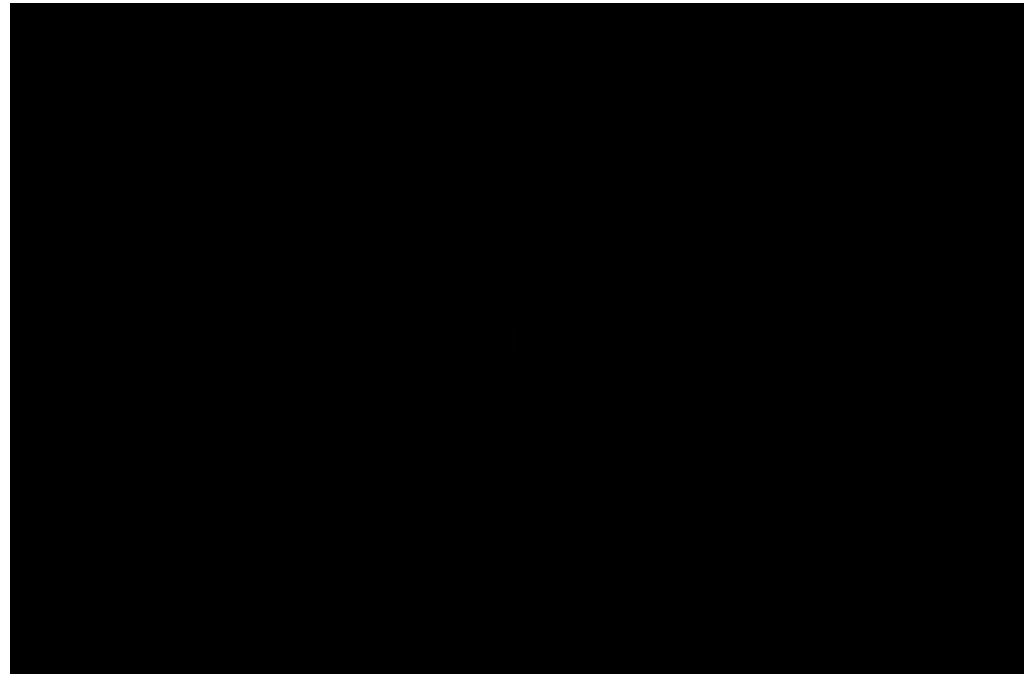
Tanh Function

Tanh Function:

Tanh function is similar to sigmoid function but its output varies from +1 to -1 and is centered at zero. The derivative of Tanh function is Stepper than of sigmoid.

Equation:

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$



Drawbacks :

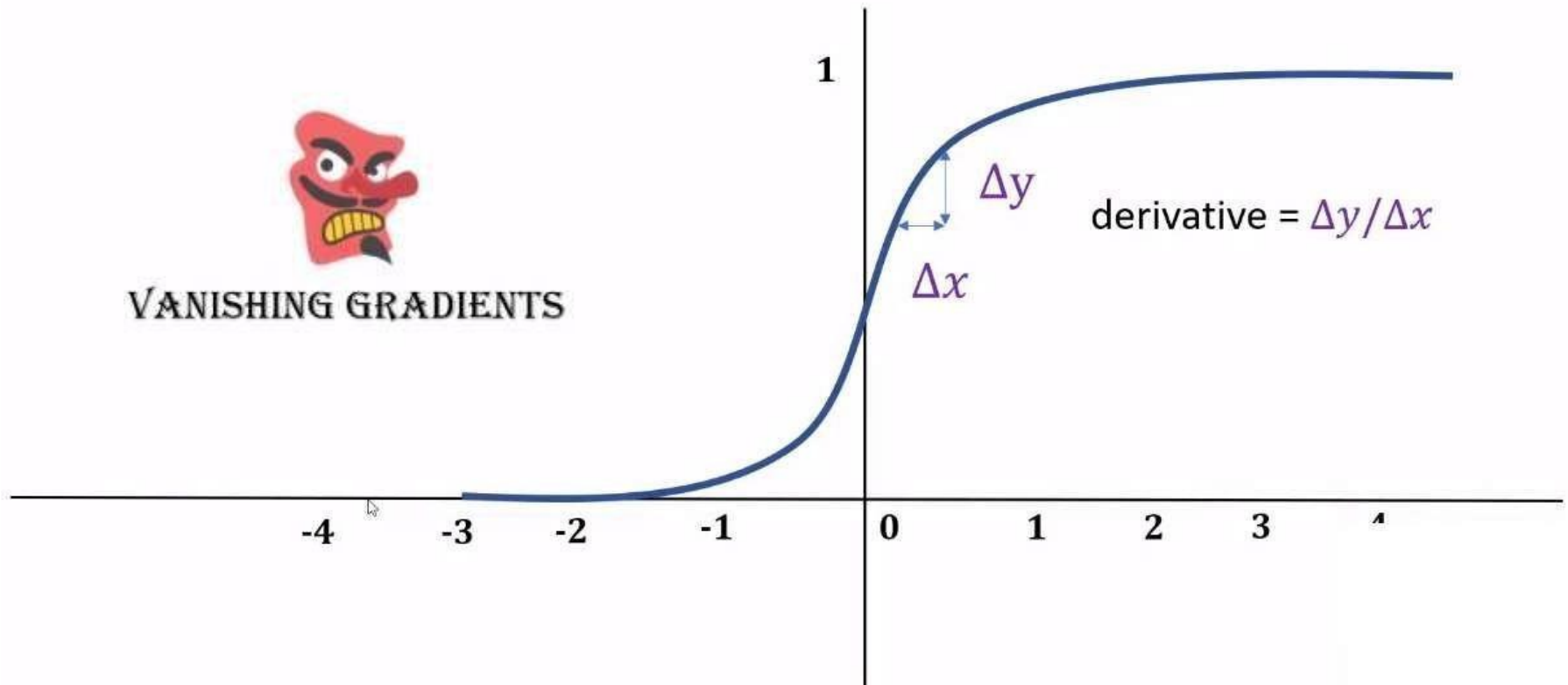
Tanh activation function also has Vanishing Gradient Problem like of Sigmoid had.



Issues with sigmoid and tanh



VANISHING GRADIENTS



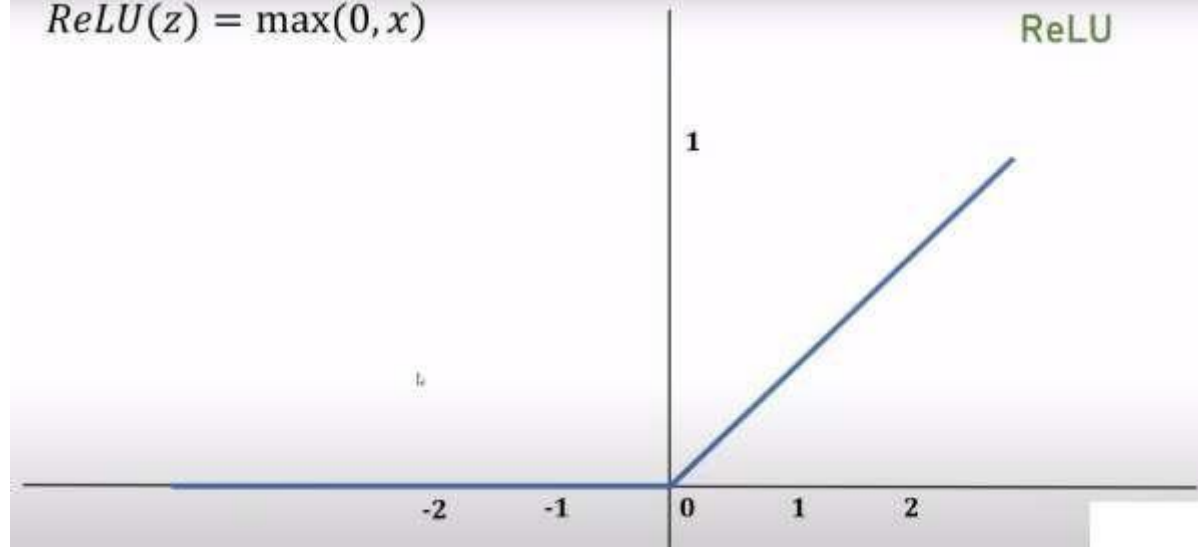
$$Dy/Dx = 0/1 = 0 \quad \square \text{ Vanishing Gradients}$$



ReLU Function

Any positive
value input
you receive
same value as
output

$$\text{ReLU}(z) = \max(0, x)$$



Drawbacks:

1. The outputs are not zero centered similar to the sigmoid activation function.
2. ReLU also have vanishing gradient problem.

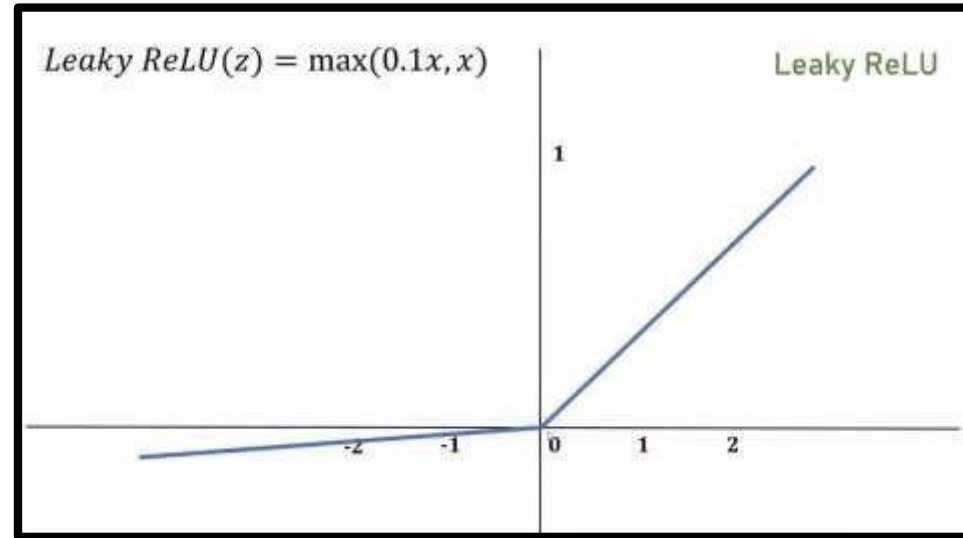


Leaky ReLU Function

Leaky ReLU Function : Leaky ReLU function is nothing but an improved version of the ReLU function. As we saw that for the ReLU function, the gradient is 0 for $x < 0$, which would deactivate the neurons in that region. Leaky ReLU is defined to address this problem. Instead of defining the Relu function as 0 for negative values of x , we define it as an extremely small linear component of x .

Equation:

$$f(x) = \begin{cases} 0.01x & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$$



Drawbacks:

There is a small tiny bit of extension in the negative area instead of a zero. This means even derivative would be a very small fraction and not zero. This ensures that learning of the neuron does not stop during backpropagation.

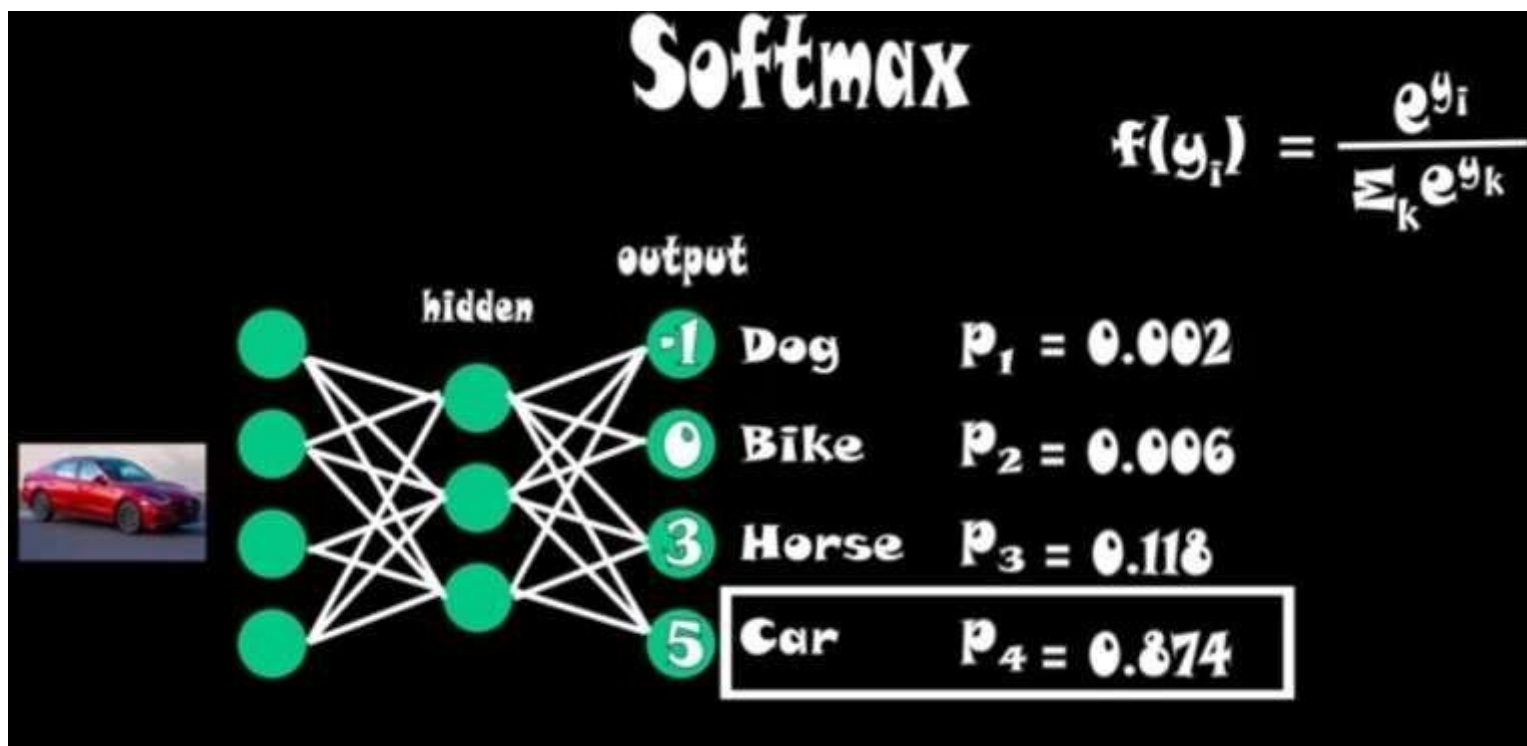


Softmax Function

$$\sigma(\vec{z})_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

Softmax Function :

1. Used for multi class classification.
2. Softmax function converts real values into probabilities.
3. It only used as output layer of neural network.
4. Higher probability as actual output.



Activation Function (Summary)

Summary of Activation Function :

- Activation functions are important because they add non-linearity in the neural network which helps the network to learn the complex relationships between real-world data.
- Sigmoid functions and their combinations generally work better in the case of classifiers.
- Sigmoid and Tanh functions should be avoided in hidden layers as it might lead to a vanishing gradient problem.
- Sigmoid and softmax functions are now generally used in the output layer in case of binary and multi-class classification respectively.
- ReLU is now popularly used in the hidden layer as it is computationally faster and does not suffer from vanishing gradient problems.
- ReLU can suffer from dying neurons problem, to avoid this Leaky ReLU is used sometimes.



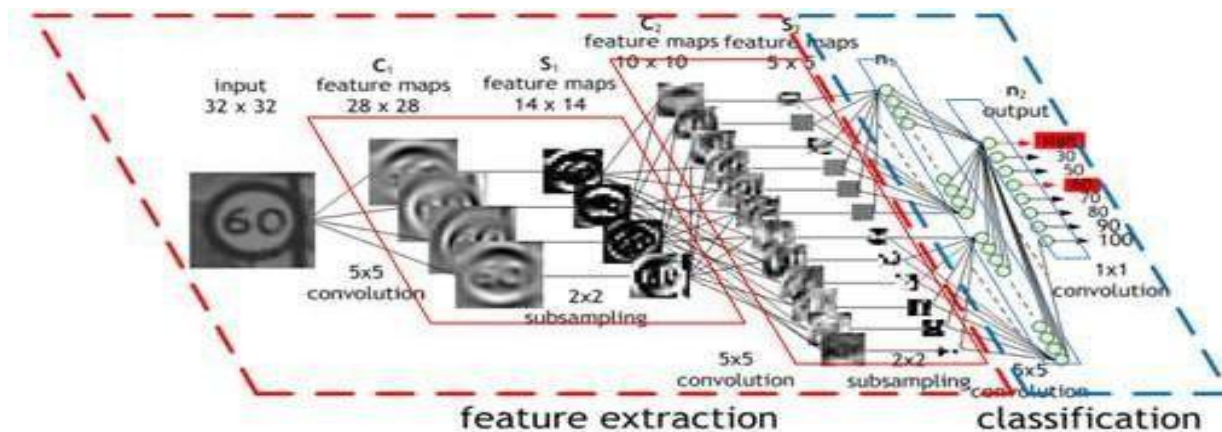
Convolutional Neural Network (CNN)

What is CNN?

A Convolutional Neural Network (CNN) is comprised of one or more convolutional layers (often with a subsampling step) and then followed by one or more fully connected layers as in a standard multilayer neural network. This is achieved with local connections and tied weights followed by some form of pooling which results in translation invariant features.

A CNN has mainly 4 layers:

- Convolutional layers
- ReLU layers
- Pooling layers
- a Fully connected layer

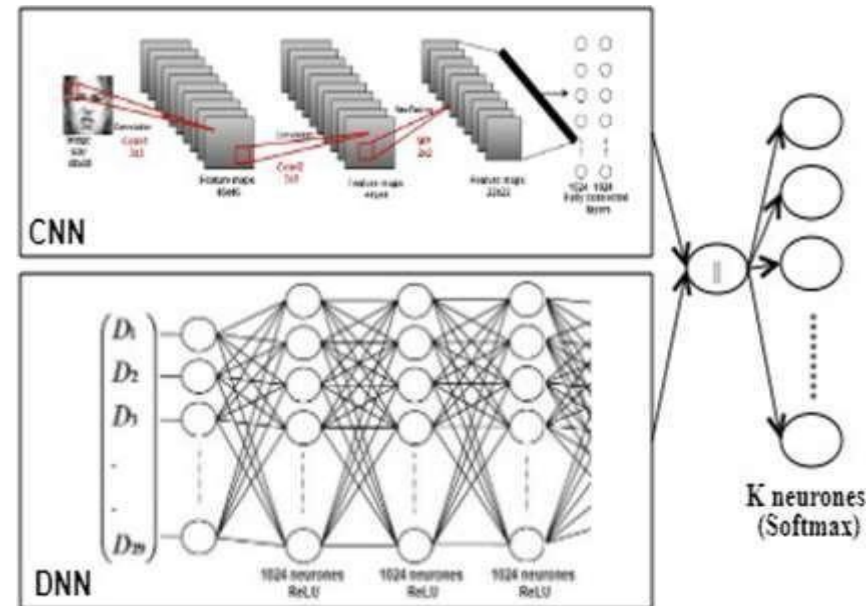
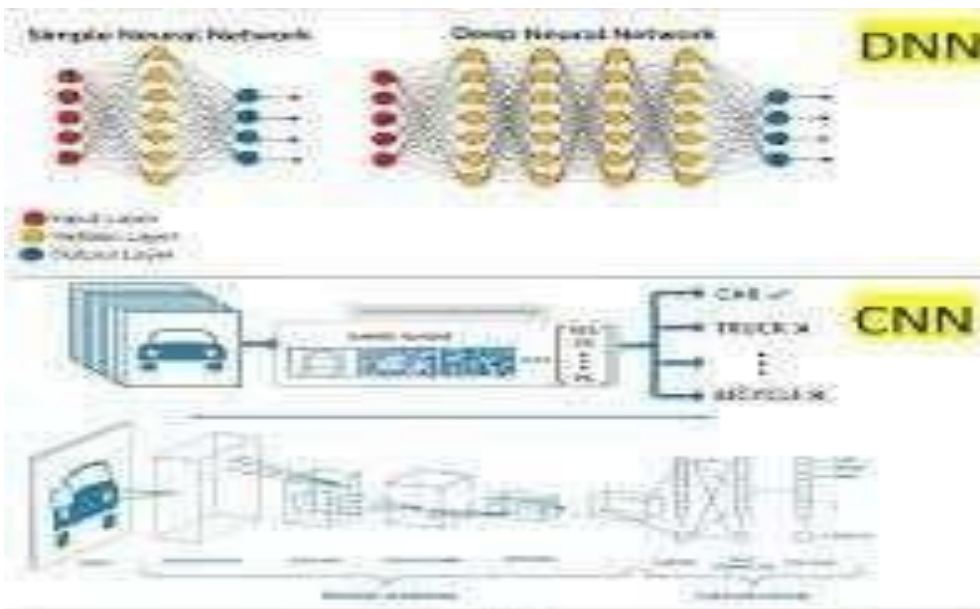


CNN VS DNN

CNN Vs DNN:

The term Deep Neural Network(DNN) refers to any neural network with several hidden layers. It can be CNN, or just a plain multilayer perceptron.

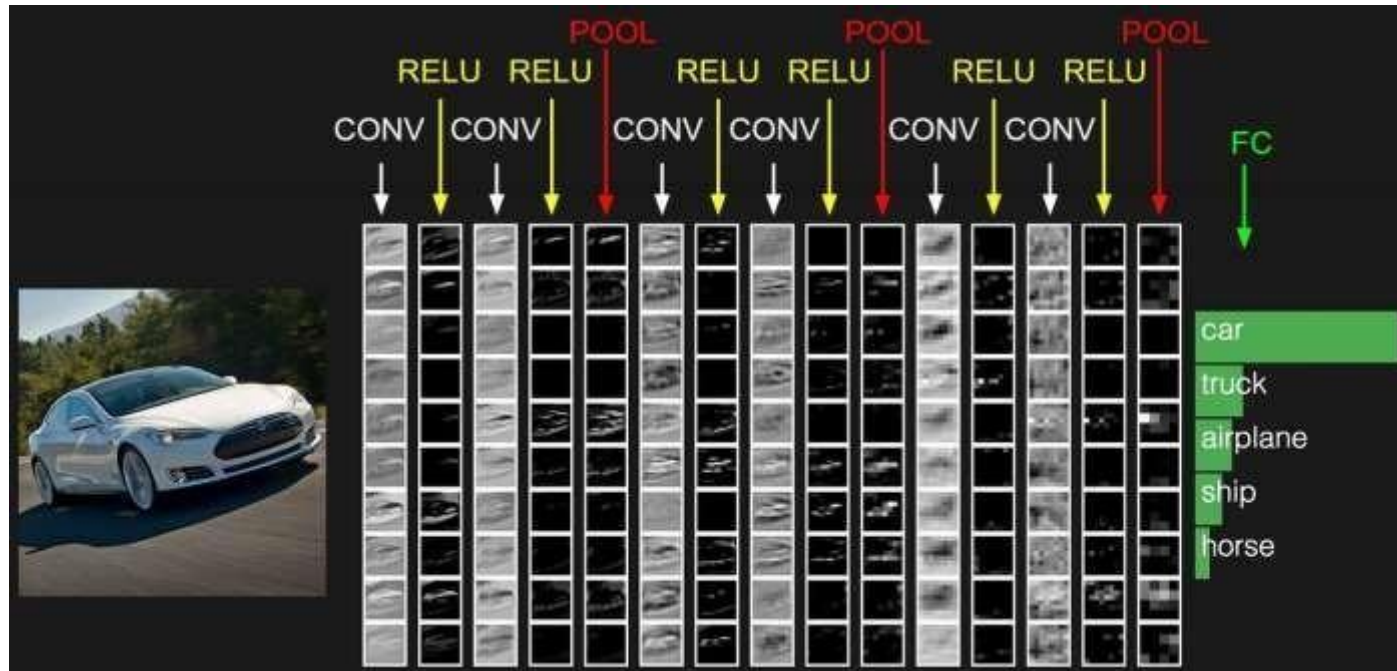
Convolutional Neural Network(CNN), is a neural network using convolution layer and pooling layer. The convolution layer convolves an area, or a stuck of elements in input data, into smaller area to extract feature. It is done by filtering an area, which is as same as to multiplying weights to an input data. The pooling layer picks an data with the highest value within an area. These layers act to extract an important feature from the input for classification.



A Typical CNN Architecture

A classic CNN architecture would look something like this:

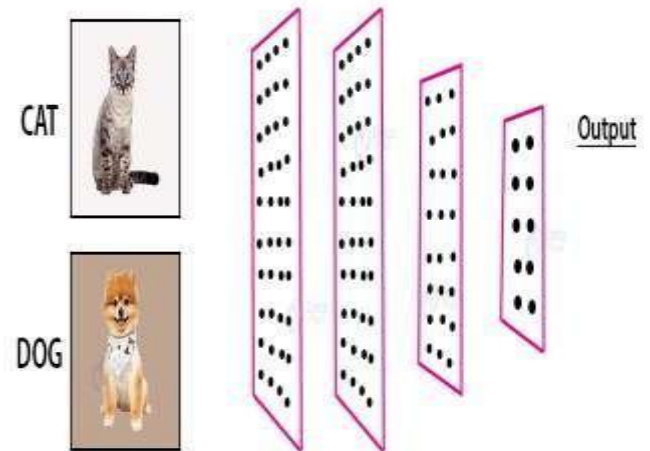
Input -> Convolution -> ReLU -> Convolution -> ReLU -> Pooling -> ReLU -> Convolution -> ReLU -> Pooling -> Fully Connected



How CNN

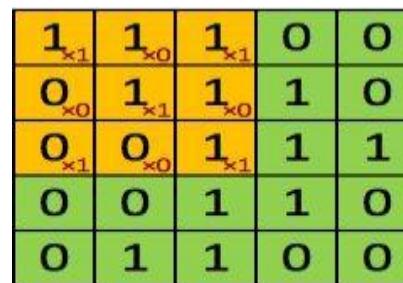
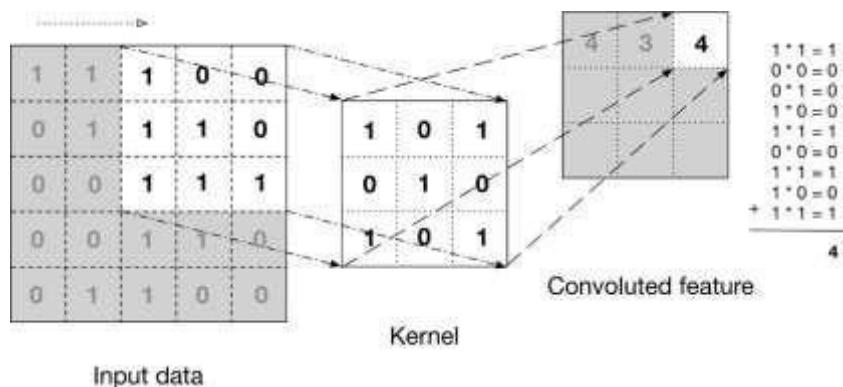
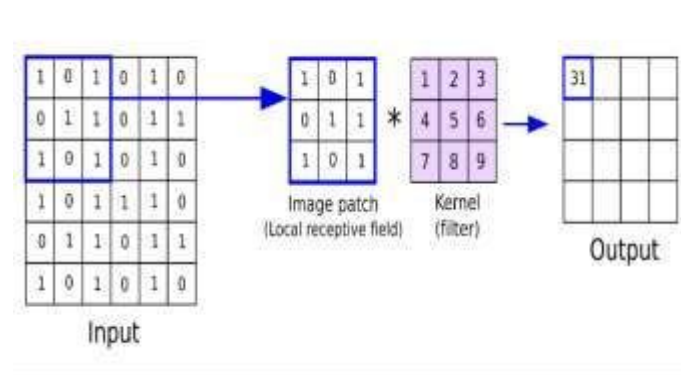
Works

Convolutional neural networks are composed of multiple layers of artificial neurons. The first layer usually extracts basic features such as horizontal or diagonal edges. This output is passed on to the next layer which detects more complex features such as corners or combinational edges. As we move deeper into the network it can identify even more complex features such as objects, faces, etc.

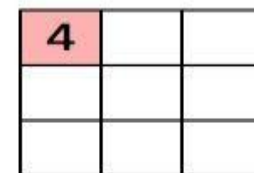


CNN Layers Definition

1. Convolutional Layer : The convolution layer (CONV) uses filters that perform convolution operations while scanning the input image with respect to its dimensions. Convolution is a linear operation involving the multiplication of a set of weights with the input images represented by metrics similar to traditional neural networks . The resulting output (O) is called the **feature map or activation map** and has all the features computed using the input layers and filters.



Image

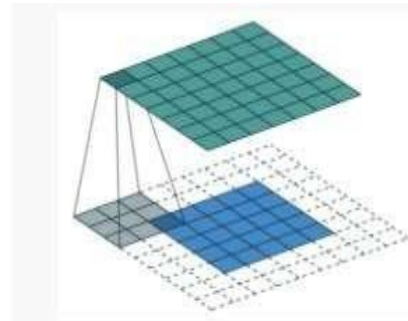


Convolved Feature

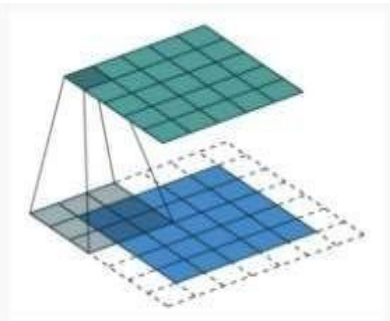
Fig: Convolution for RGB image

CNN Layers Definition(Continued)

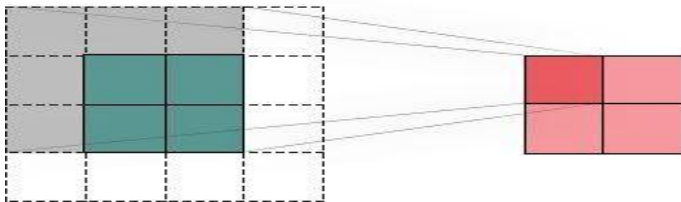
2. Padding : It refers to the amount of pixels added to an image when it is being processed by the kernel of a CNN. For example, if the padding in a CNN is set to zero, then every pixel value that is added will be of value zero..



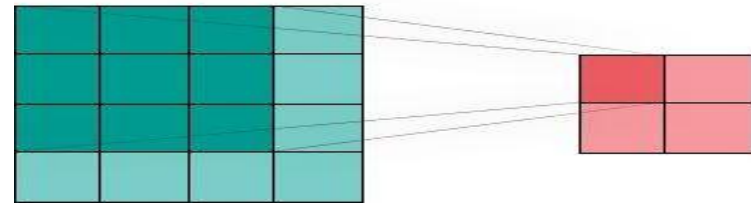
Full padding. Introduces zeros such that all pixels are visited the same amount of times by the filter. Increases size of output.



Same padding. Ensures that the output has the same size as the input.



Same padding



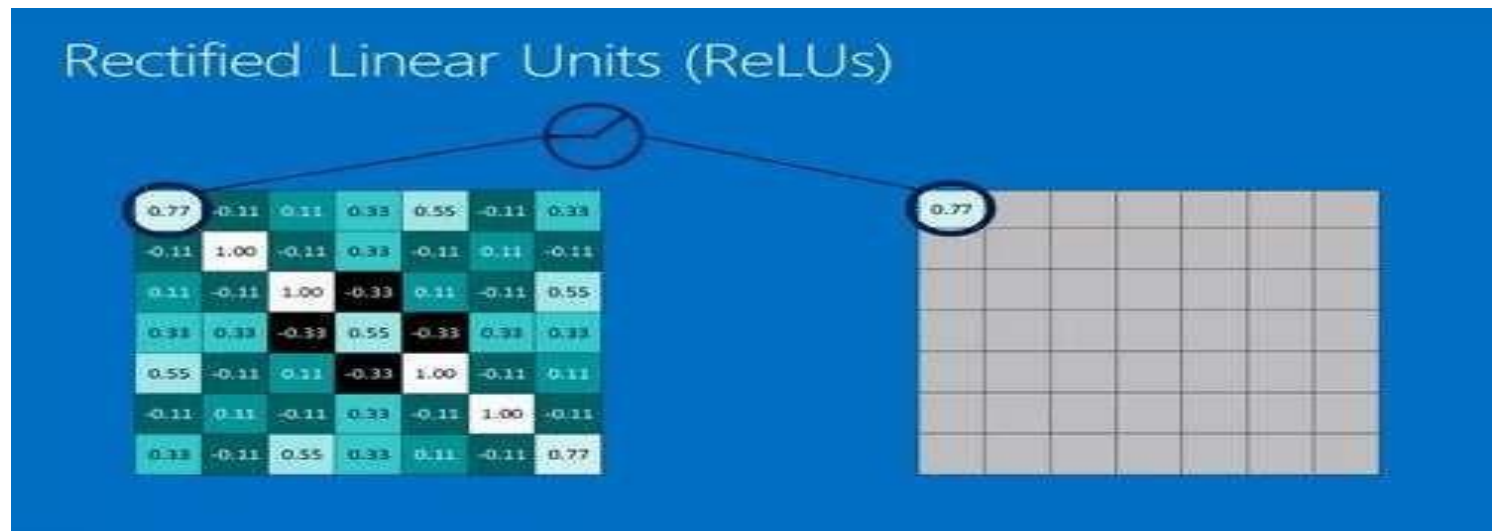
Valid padding



CNN Layers Definition(Continued)

3. ReLU layer:

The ReLU (rectified linear unit) layer is another step to our convolution layer. The purpose of applying the rectifier function is to increase the non-linearity in our images. The reason we want to do that is that images are naturally non-linear. It removes negative values from an activation map by setting them to zero. The rectifier serves to break up the linearity even further in order to make up for the linearity that we might impose an image when we put it through the convolution operation.

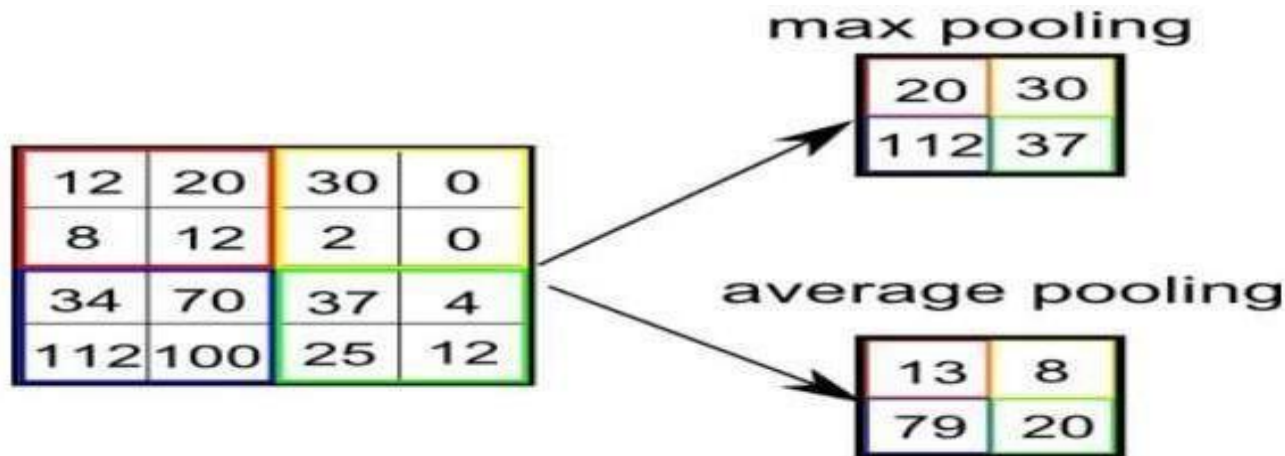


CNN Layers Definition(Continued)

4. Pooling layer: The Pooling layer is responsible for reducing the spatial size of the Convolved Feature. This is to decrease the computational power required to process the data by reducing the dimensions. There are two types of pooling average pooling and max pooling. **Max Pooling** find the maximum value of a pixel from a portion of the image covered by the kernel. **Average Pooling** returns the average of all the values from the portion of the image covered by the Kernel.

3.0	3.0	3.0
3.0	3.0	3.0
3.0	2.0	3.0

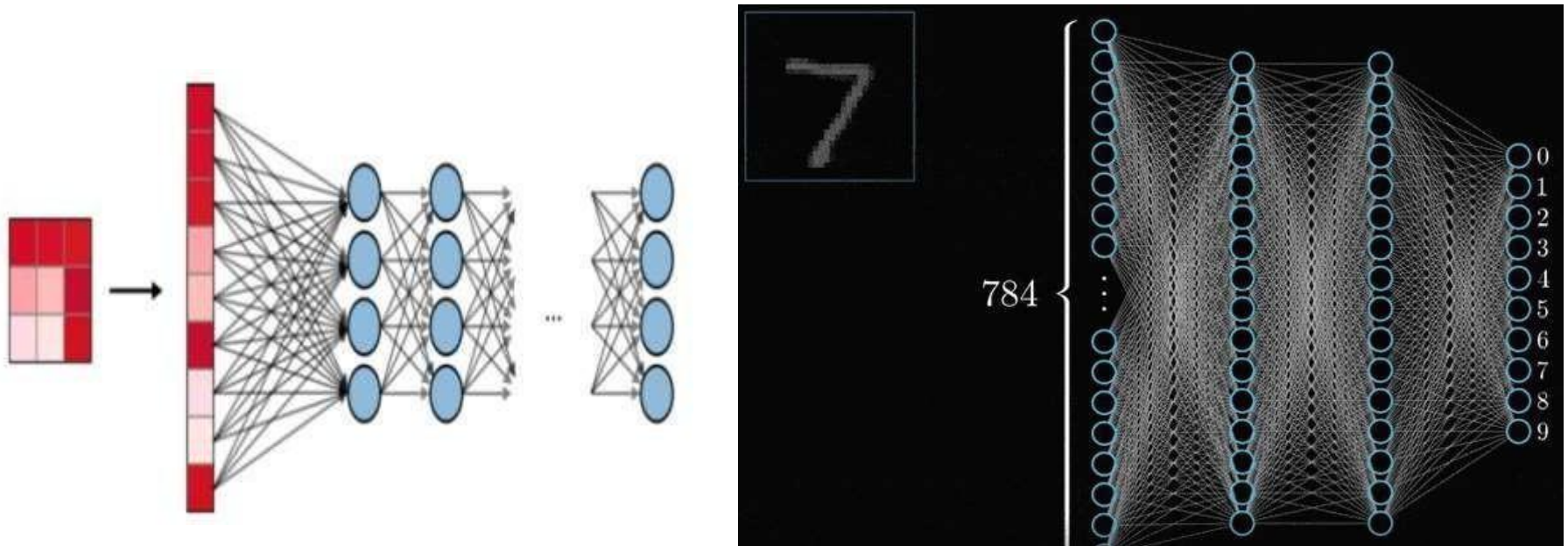
3	3	2	1	0
0	0	1	3	1
3	1	2	2	3
2	0	0	2	2
2	0	0	0	1



CNN Layers Definition(Continued)

5. Fully connected layer :

At this step, we add an **artificial neural network** to our convolutional neural network. **The fully connected layer (FC) operates on a flattened input where each input is connected to all the neurons.** These are usually used at the end of the network to connect the hidden layers to the output layer, **which helps in optimizing the class scores.**



Fully connected layer



The End

**Thank
You**

